



NITI Aayog



SECTOR AT A GLANCE 11

Oil and Natural Gas Energy in India

Energy in India

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OIL AND NATURAL GAS ENERGY IN INDIA

ENERGY IN INDIA

The **'Sector-at-a-Glance'** provides an easy-to-read overview and basic understanding of a sector, sub-sector, or thematic area. The facts and insights on the sector are supported by relevant data and best practices from credible government and multilateral sources. The document covers the structure, key trends, opportunities and challenges, government initiatives, and the way forward for the sector.

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Overview

Oil and gas are vital cogs in India's energy supply landscape and the vast majority of what we use is imported. India's oil refinery capacity has increased by 18% over the last decade, making it the world's fourth-largest crude oil refining hub as of 2023.¹ India is unable to meet its oil demand through domestic production and imports most of its crude oil (87%), mainly from Iraq, Saudi Arabia, and the UAE.² Further, India has 1138.67 billion cubic meters (BCM) of natural gas reserves as per provisional 2021-22 numbers, which have increased marginally from 2016-17 levels. As of 2022-23, India also had an import dependency of about 43.9% concerning natural gas consumption, and imports mainly from Qatar, UAE, USA, Oman, and Australia.³

Oil and natural gas are expected to remain critical to our future needs. Oil and petroleum products will contribute about 25% (724.46 mtoe) of India's total energy supply by 2047, and natural gas will contribute ~6% (176.92 mtoe).⁴ Effective and sustainable adoption of oil and natural gas can lead to increasing domestic production to meet short-term needs, greater usage of carbon capture and abatement technologies, and support to industry players to enable transition to clean energy.

Despite progress, challenges which persist include high import dependency, inadequate pipeline coverage, transportation, and marketing, unreliable and poor quality of supply in rural areas, and insufficient adoption of emission mitigation technologies.

Given these opportunities and challenges, oil and natural gas resource exploration, supply, pricing, emission mitigation, and transition to clean energy are the major policy focus areas. Several ongoing and recent central government initiatives, including but not limited to Hydrocarbon Exploration and Licensing Policy, Coal Bed Methane Policy, and Pradhan Mantri Ujjwala Yojana aim to increase exploration and production efforts, and improve supply especially for disadvantaged groups. These efforts along with several state- and local-level policies and initiatives, seek to establish sustainable energy resource use and energy security in India.

Efforts to promote exploration and diversify production to increase self-reliance, improve the efficiency and environmental sustainability of oil production and transport, enable a sustainable workforce transition, and support integrated planning will be critical to continue advancing India's oil and gas sector. This note presents key trends related to different components of oil and gas in India and actions for the way forward.

Figure 1: Key oil and natural gas energy statistics

236.57 Mtoe

Share of oil and petroleum products in India's energy supply in 2022

62 Mtoe

Share of natural gas in India's energy supply in 2022

724.46 Mtoe

Estimated share of oil and petroleum products in India's energy supply in 2047

176.92 Mtoe

Estimated share of natural gas in India's energy supply in 2047

¹ Ministry of Statistics and Programme Implementation, Energy – Statistical Year Book India, 2015; Ministry of Statistics and Programme Implementation, Energy Statistics India, 2023; Ministry of Petroleum and Natural Gas, History and Evolution, Accessed June 2023

² Petroleum Planning and Analysis Cell, Snapshot of India's Oil & Gas data, 2023; Ministry of Mines, Indian Minerals Yearbook 2020, 2022

³ Petroleum Planning and Analysis Cell, Snapshot of India's Oil & Gas data, 2023; Ministry of Petroleum and Natural Gas, Indian Petroleum & Natural Gas Statistics, 2021-22; Department of Commerce, Import: Commodity-wise all countries: Liquefied Natural Gas, 2023

⁴ NITI Aayog, India Energy Security Scenarios 2047, Accessed September 2023

29.2 MMT

Crude Oil Production 2022-23

34.5 BCM

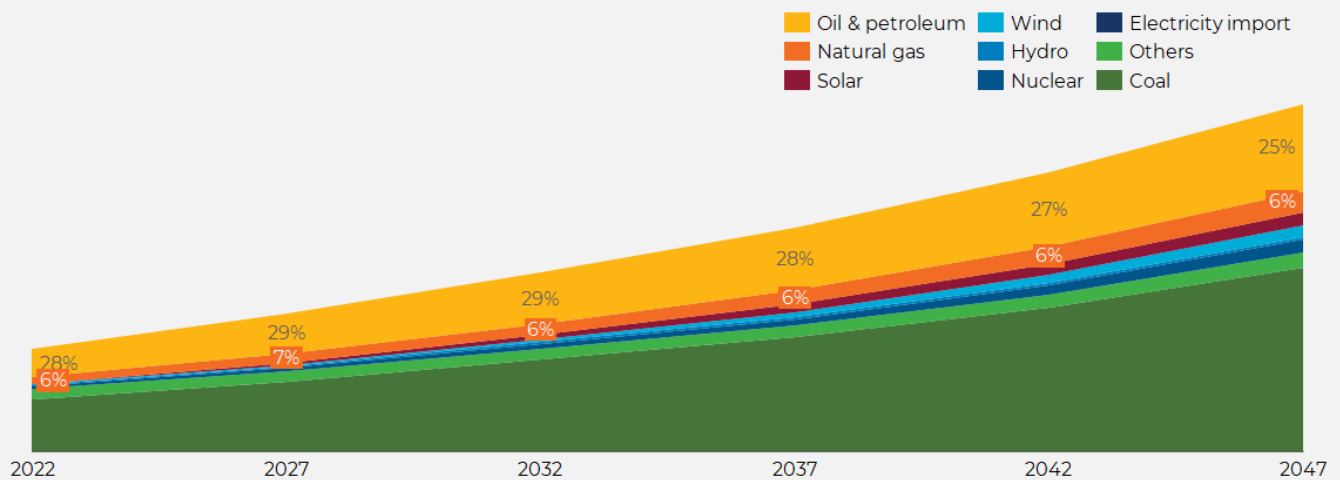
Natural Gas Production 2022-23

87.4%

India's import dependency for crude oil in 2022-23*

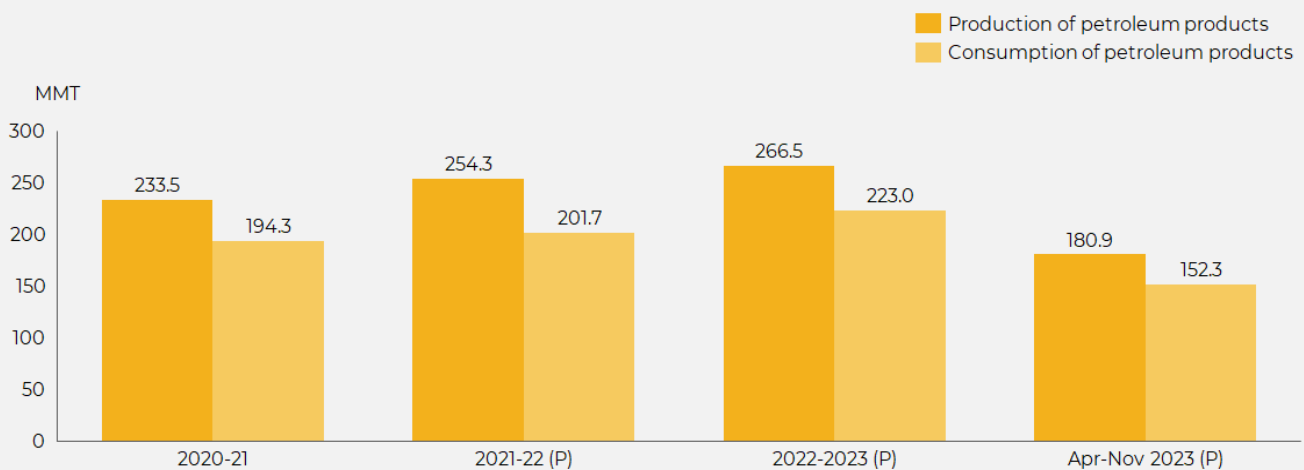
Source: NITI Aayog, India Energy Security Scenarios 2047, Accessed December 2023; Ministry of Petroleum and Natural Gas, India Hydrocarbon Outlook, 2023; Ministry of Petroleum and Natural Gas, Snapshot of India's Oil and Gas Data - Monthly Ready Reckoner - November, 2023
Notes: * On POL consumption basis

Figure 2: Estimated growth of oil and petroleum products, and natural gas in India's energy supply between 2022 and 2047



Source: NITI Aayog, India Energy Security Scenarios 2047, Accessed December 2023

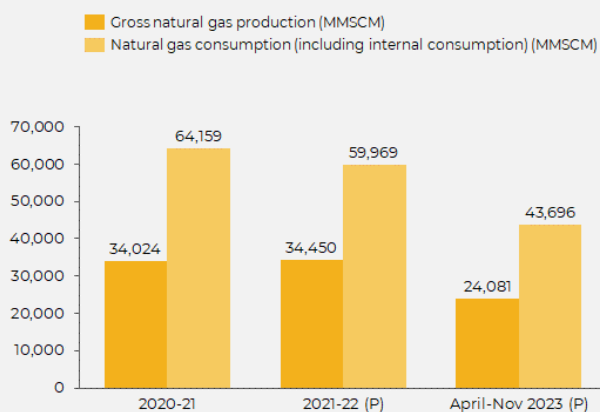
Figure 3: Production & Consumption of Petroleum Products, in million metric tonnes



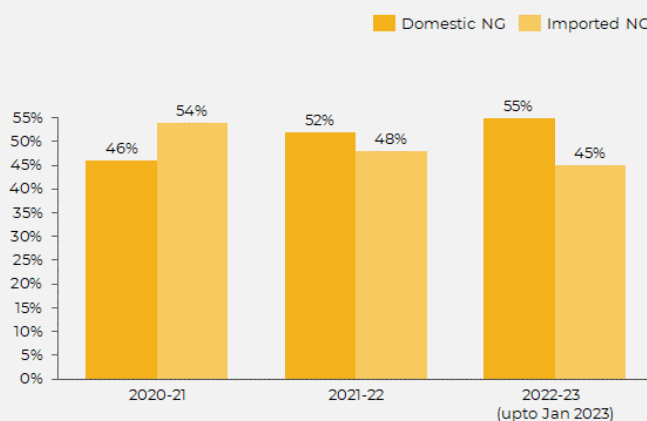
Source: Ministry of Petroleum and Natural Gas, Snapshot of India's Oil and Gas Data - Monthly Ready Reckoner - November, 2023

Figure 4: Production and consumption of natural gas and domestic and imported natural gas share in India in %

Production and consumption of natural gas (MMSCM)

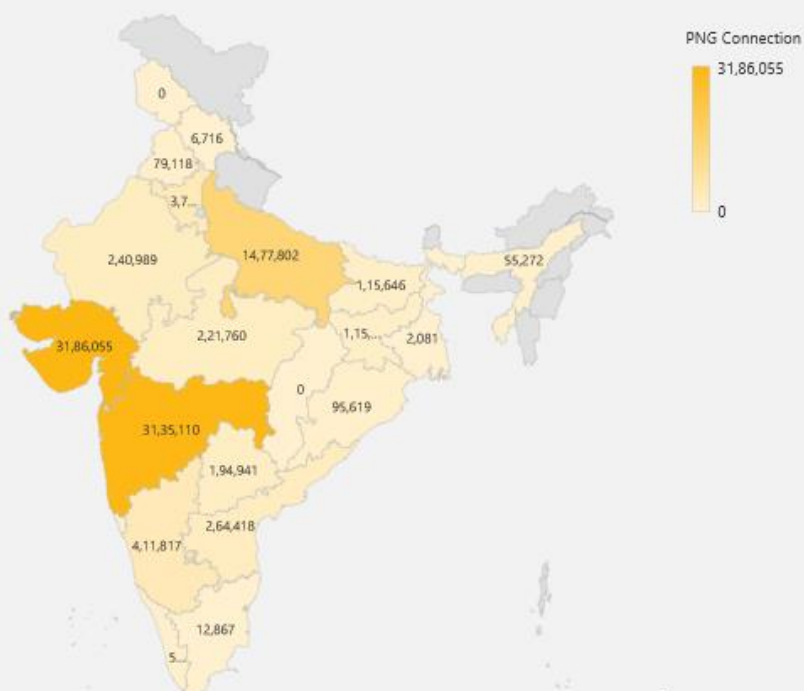


Domestic and imported natural gas share in India (%)



Source: Ministry of Petroleum and Natural Gas, Snapshot of India's Oil and Gas Data - Monthly Ready Reckoner - November, 2023

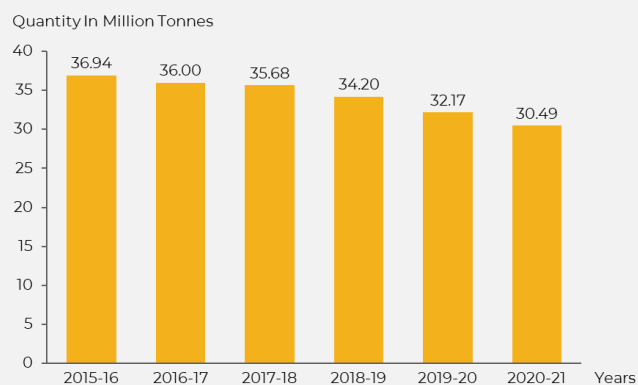
Figure 5: State-wise PNG connections and CNG stations across India as of 31.10.2023



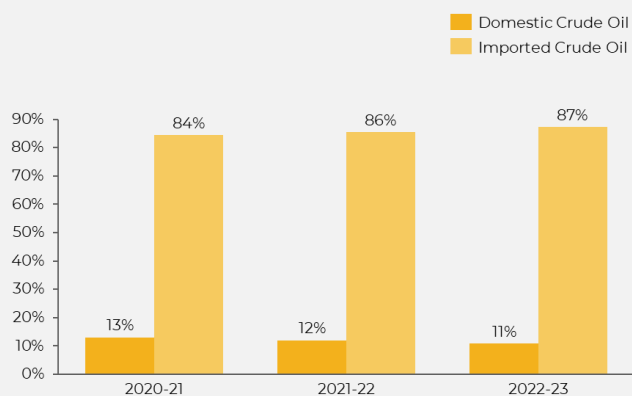
Source: Ministry of Petroleum and Natural Gas, Snapshot of India's Oil and Gas Data - Monthly Ready Reckoner - November, 2023

Figure 6: Production of crude oil & condensate, 2015-16 to 2020-21 and domestic and imported crude oil share in India in %, 2020-21—2022-23

Production of crude oil & condensate, 2015-16 to 2020-21



Domestic and Imported Crude Oil share in India (%)



Source: Ministry of Mines, Indian Minerals Yearbook 2021, 2023; Ministry of Petroleum and Natural Gas; PPAC; Petroleum Planning and Analysis Cell, India's Oil and Gas Ready Reckoner, 2023

Opportunities and challenges

Opportunities

This section highlights the outcomes policymakers can achieve by adopting effective oil and natural gas practices. These revolve primarily around the potential for improved domestic production of oil and natural gas, increase in the use of carbon abatement technologies and supporting the oil and natural gas sector's transition to clean energy.



Increasing domestic production to meet short-term needs:

More than a third of India's current energy demand is met through hydrocarbons.⁵ Almost half of the sedimentary basins in India have undiscovered hydrocarbon potential.⁶ The Indian government is thus implementing various policies like Hydrocarbon Exploration and Licensing Policy, Discovered Small Field Policy, and those under the Production Sharing Contract (PSC) regime to increase domestic production of oil and natural gas and reduce India's import dependency.⁷



Using carbon capture and abatement technologies:

Capturing and abating carbon emissions is critical to our energy transition goals, especially in sectors like oil and natural gas. Technologies like carbon capture, utilisation, and storage (CCUS) present the dual opportunity of helping reduce atmospheric emissions and potential use in other activities like enhancing oil recovery from mature oil fields.⁸ India has already started exploring the opportunity to use CO₂ for enhanced oil recovery, and one of India's first CCUS projects is being led by Indian Oil Corporation Ltd (IOCL) and Oil and Natural Gas Corporation (ONGC), who are setting up India's first large-scale carbon capture project in Gujarat.⁹ Furthermore, to leverage India's vast potential in this carbon abatement space, two National Centres of Excellence in Carbon Capture and Utilization are being established at IIT Bombay and JNCASR, Bengaluru, which will act as multi-disciplinary, research, design, and capacity-building hubs for high-quality research, and related applications in the field of CCU.¹⁰



Supporting oil and natural gas players in their transition to clean energy:

India's oil and natural gas giants are taking several steps in line with India's net zero targets and can become inspiring examples for the global economy. For instance, Indian Oil has committed to achieve net-zero operational emissions by the year 2046.¹¹ Another notable example is from the carbon emission management practices followed by ONGC. The company joined hands with the United States Environmental Protection Agency (USEPA) in 2007 to reduce methane emissions by employing emission reduction technologies and conducts regular surveys to guard against gas leaks at production sites.¹² Other oil and gas players, such as Bharat Petroleum Corporation Limited (BPCL), Hindustan Petroleum Corporation Limited (HPCL) and GAIL India Limited have all announced their Net Zero Commitments to be achieved by 2040.¹³

⁵ Ministry of Mines, Indian Minerals Yearbook 2021, 2023

⁶ Ministry of Mines, Indian Minerals Yearbook 2021, 2023

⁷ Ministry of Petroleum and Natural Gas, Various steps have been taken by the Government to reduce import of Crude Oil, 2022

⁸ NITI Aayog, Carbon Capture Utilization and Storage (CCUS)- Policy Framework and Deployment Mechanism in India, 2022; Federation of Indian Petroleum Industry, Journal: Voice of Indian Oil & Gas Industry, 2021

⁹ Ministry of Petroleum, Oil and Natural Gas, Draft 2030 Roadmap for Carbon Capture Utilization and Storage (CCUS) for Upstream E&P Companies, 2022

¹⁰ Ministry of Science & Technology, India to have two National Centres of Excellence in Carbon Capture & Utilization at IIT Bombay & JNCASR, Bengaluru, supported by DST, 2022

¹¹ IndianOil, IndianOil to go Net Zero by 2046, accessed July 2023

¹² Oil and Natural Gas Corporation Limited, Carbon Management, accessed July 2023

¹³ BPCL, What is Carbon Neutrality and How can we Achieve it by 2040, 2023; Hindustan Petroleum Corporation Limited (HPCL); Sustainability Report, 2022; GAIL India Limited, GAIL's Net-Zero Action Plan, Accessed December 2023

Challenges

This section highlights some of the major challenges and constraints facing oil and natural gas energy in India. Efforts to resolve such challenges will ensure enhanced pipeline coverage, improved supply in rural areas and better adoption of mitigation technologies.



High import dependence:

The crude oil production in 2012-13 was 37.71 million metric tonnes (MMT) which declined to 29.7 MMT in 2021-22 driven by a decline in existing reserves, ageing fields, and lack of new oil fields.¹⁴ Similarly, the natural gas production in 2012-13 was 40.68 billion cubic meters (BCM) which declined to 34.02 BCM in 2021-22, driven by old and ageing fields. As a result, oil and natural gas import has steadily risen during this period. Crude oil import in 2012-13 was 82% of total oil demand, which increased to 86% in 2021-22. Similarly, the import of natural gas has risen from 30% of the total demand for gas in 2012-13 to 47% in 2021-22.¹⁵



Inadequate pipeline coverage, transportation, and marketing:

Southern and eastern regions of India lack sufficient pipeline coverage, and land acquisition challenges have caused delays in major pipeline projects like the Kochi-Mangalore-Bangalore pipeline. Moreover, infrastructure such as CNG filling stations and city gas networks for domestic use is limited to a few cities.¹⁶ Existing transportation infrastructure also faces several shortcomings as highlighted by a government report which observed that oil marketing companies in India incurred high extra costs due to variations regarding planned logistics. Some reasons for these deviations were a mismatch in supply from refineries and the non-availability of planned modes of transport, among others. The absence of data regarding reasons for these logistics deviations further prevents corrective action.¹⁷



Unreliable and poor quality of supply in rural areas:

Usage of firewood and kerosene are still prevalent for domestic cooking in rural areas.¹⁸ Further, high prices of LPG restrict sustained use, especially by poor households. Additionally, vast differences exist in energy use and service quality between urban and rural areas. For instance, as part of the Pradhan Mantri Ujjwala Yojana (PMUY) scheme which seeks to make clean cooking fuel available to rural households, LPG distributors did not focus on training regarding safety measures.¹⁹ As a result, many beneficiaries were reported as following unsafe LPG practices, which could have led to fatal accidents.²⁰



Insufficient adoption of emission mitigation technologies:

India's growth in the oil and gas sector has led to a parallel rise in hydrocarbon emissions.²¹ India has the potential to utilise carbon abatement technologies like CCUS for functions like oil recovery, but it lacks natural CO₂ sources for this process.²² Moving to carbon abatement technologies like CCUS is typically very expensive, and CCUS costs remain high despite extensive research on the need for alternative financial mechanisms. Big oil and natural gas companies require significant capital investment and advanced technologies to achieve their net-zero emissions targets. For the use of this technology at scale, India would require suitable

¹⁴ Petroleum Planning and Analysis Cell, India's Oil and Gas Ready Reckoner, 2023

¹⁵ Ministry of Petroleum and Natural Gas, Indian Petroleum and Natural Gas Statistics, 2022; Ministry of Statistics and Programme Implementation, Energy Statistics India, 2023

¹⁶ Council on Foreign Relations, What Will It Take to Turn Natural Gas Around in India?, 2016

¹⁷ Ministry of Petroleum & Natural Gas, Supply Logistics Operations of MS, HSD and LPG in Oil Marketing Companies, 2022

¹⁸ Ministry of Petroleum and Natural Gas, Subsidy on Petroleum, 2023

¹⁹ CEEW, State of Clean Cooking Energy Access in India, 2020

²⁰ CEEW, State of Clean Cooking Energy Access in India, 2020

²¹ Ministry of Petroleum and Natural Gas, The Green Shift, 2023

²² Federation of Indian Petroleum Industry, Journal: Voice of Indian Oil & Gas Industry, 2021

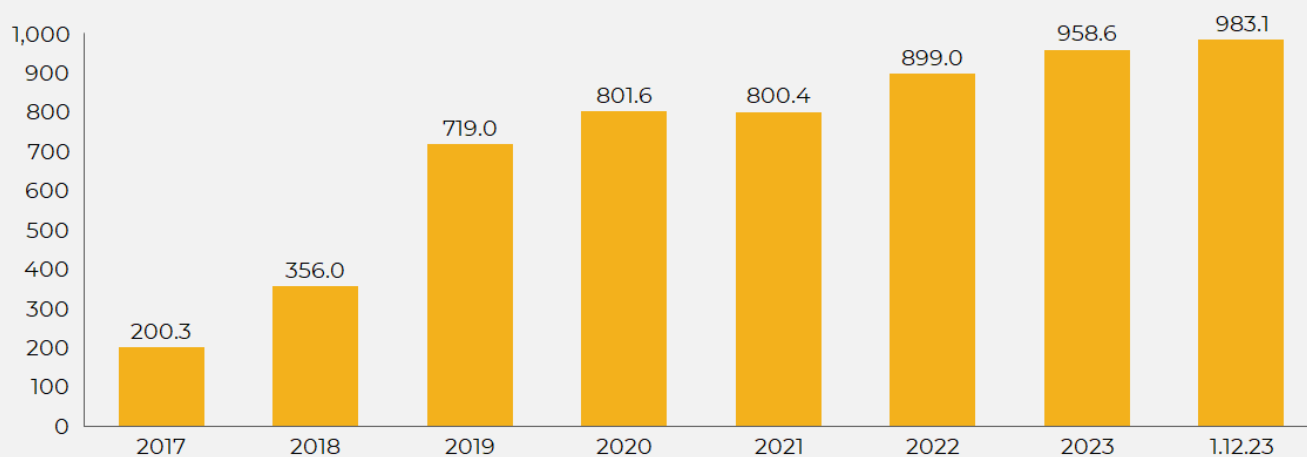
infrastructure and policy assurances that can help reduce technology costs.²³ It would also need to conduct technology transfers and seek finance from developed economies.



Scope to expand assessments of sedimentary basins

India's sedimentary basin cover spans 3.4 million square kilometres, out of which 61% has been appraised through geoscientific surveys, with 39% yet to be assessed.²⁴

Figure 7: Number of PMUY beneficiaries (in lakhs)



Source: Ministry of Petroleum and Natural Gas, Snapshot of India's Oil and Gas Data - Monthly Ready Reckoner - November, 2023

²³ Indian Council for Research on International Economic Relations (ICRIER), Exploring Carbon Capture, Utilisation and Storage in the Indian Context: ICRIER G20 Policy Series - Run Up to 2023, 2022

²⁴ Ministry of Petroleum and Natural Gas, India's Hydrocarbon Outlook, 2023

Institutional framework and policy landscape –

Key ministries and departments

This section highlights the major central government bodies and national-level institutions that oversee oil and natural gas energy in India.



Ministry of Power

Administers, formulates, and implements power sector-related policies and provides R&D and technical assistance relating to hydroelectric and thermal power, transmission system networks and distribution systems in the States/UTs



Ministry of New and Renewable Energy (MNRE)

Develops renewable and clean energy technologies, systems, products, and services in line with global standards and deploys these for energy applications for sectors like power and transport. Concerned with oil and natural gas, it works towards development and deployment of alternate fuels like hydrogen, bio-fuels and synthetic fuels and their applications to contribute towards bridging the gap between domestic oil supply and demand, in turn enabling lesser dependency on oil imports and fostering energy security.



Ministry of Petroleum and Natural Gas

Undertakes exploration and extraction of oil and natural gas, as well as the refining, distribution, and marketing of petroleum products, alongside handling activities related to the import, export, and conservation of these products



NITI Aayog

Conducts policy analysis of energy sectors related to power, coal, petroleum and natural gas, new and renewable energy, and atomic power; and undertakes steps to improve energy planning, and data management in India



Central Electricity Authority

Acts as a technical support base for power sector stakeholders and supports policy formulation, establishing technical standards, project monitoring and human resource skilling



Petroleum Planning and Analysis Cell

Adopts the latest technology for strengthening data systems in oil and gas sector, assists in oil and gas pricing and administration of schemes, and analyses domestic and international energy markets



Petroleum and Natural Gas Regulatory Board

Facilitates investment flow into infrastructure for efficient transportation and distribution of petroleum and natural gas and promotes fair trade practices to ensure enhanced competitiveness of Indian markets and customer satisfaction

Directorate General of Hydrocarbon

Implements policies and contracts concerning exploration of coal, oil, and natural gas, promote investments, conduct monitoring of exploration and production activities alongside the development of non-conventional hydrocarbon energy sources

Major policies and programs

This section highlights existing policies related to oil and natural gas energy and their most prominent features. A more comprehensive list of policies is provided in the policy repository of the NITI Knowledge Platform.



Hydrocarbon Exploration and Licensing Policy (HELP), 2016

HELP seeks to increase domestic oil and natural gas production in India and reduce import dependency by enabling greater ease of doing business. It aims to provide greater freedom to industry operators for effective exploration of existing hydrocarbon resources. It also grants a uniform license to investors to explore and extract hydrocarbon resources. Under the HELP regime, several exploration activities related to unconventional hydrocarbons have been undertaken in the last five years.²⁵



Reforms in Hydrocarbon Exploration and Licensing Policy, 2019²⁶

These reforms sought to enhance exploration activities by fiscal and contractual terms, bidding of exploration blocks in Category II and III sedimentary basins without government sharing in production or revenue, facilitating quicker monetization of discoveries through extended fiscal benefits, encouraging gas production by granting freedom in marketing and pricing, incorporating cutting-edge technology and capital, granting greater operational autonomy to national oil companies for collaborations, and inviting private sector participation in enhancing production methods within nominated fields. Additionally, the reforms aimed to streamline approval procedures and promote a business-friendly environment by introducing an electronic single window mechanism.



Discovered Small Field Policy (DSF), 2015

This policy grants "discovered small fields" (land areas containing hydrocarbon resources) for international bidding to facilitate early monetisation of these untapped resources. It encompasses aspects like a revenue-sharing contract model with minimal regulatory obligations, no minimum biddable work programme or prior technical qualifications required. The DSF bid round III was initiated in 2021, offering 32 contract areas comprising 75 discoveries.²⁷



Deregulation of domestically produced crude oil sales, 2022

This government initiative guarantees marketing independence for all Exploration and Production (E&P) operators. The requirement under Production Sharing Contracts (PSCs) to sell crude oil to the government or related entities is waived off. Thus, all E&P companies will be free to sell crude oil extracted from their fields in the domestic market.²⁸

²⁵ Ministry of Petroleum and Natural Gas, Primary objective of the HELP is ease of doing business and operational freedom to Operators, 2022

²⁶ Ministry of Petroleum & Natural Gas, Import/Export of Oil, 2019

²⁷ Ministry of Petroleum and Natural Gas, Discovered Small Field Bid Round-III launched, 2021

²⁸ Ministry of Petroleum and Natural Gas, Cabinet approves Deregulation of Sale of Domestically Produced Crude Oil, 2022

Coal Bed Methane Policy, 1997

This policy aims to extract Coal Bed Methane (CBM) from coal-rich regions. Recently, the Ministry of Petroleum and Natural Gas has initiated several new reforms like frameworks for early monetisation of CBM, exploration and exploitation of unconventional hydrocarbons and promotion of enhanced recovery methods for oil and gas. As of 2020, in the four bidding rounds of CBM, 33 blocks have been allotted to various companies.²⁹

Revised Domestic Gas Pricing Guidelines, 2023³⁰

The guidelines mandate that the cost of certain natural gas blocks will be determined at 10% of the monthly average of the Indian Crude Basket and will be announced monthly. For the gas extracted by ONGC & OIL from their designated areas, the Administered Price Mechanism (APM) price will have both a floor and a ceiling. Additionally, gas obtained from new wells or interventions in ONGC & OIL's designated areas will be eligible for a 20% premium over the APM price. The guidelines aim to establish a steady pricing system for local gas consumers while also offering sufficient safeguards for producers against market volatility, along with incentives to boost production.

Gas Utilization Policy³¹

As per guidelines of this policy, domestic gas is allocated to different sectors on priority basis. The first order of priority is given to sectors such as CGD sector for PNG (Domestic) and CNG (Transport), Fertilizer, Power, LPG etc. Any demand over and above domestic gas may be met through imported Re-gasified Liquefied Natural Gas (RLNG) which is imported under open general license on mutually agreed terms.

Pradhan Mantri Ujjwala Yojana (PMUY)³²

The scheme aims to release deposit-free LPG connections in the name of adult woman members of poor households in India. As on 01.07.2023, there are close to 9.5 crore PMUY beneficiaries, out of which about 8.4 crore have taken at least one refill during 2022-23. Under Ujjwala 2.0, a special provision has been made for migrant families who can use a self-declaration instead of address proof and ration card to apply for a PMUY connection.

Additional policies and provisions to enhance the production and exploration of oil and gas in India³³

- Policy for relaxations, extensions, and clarifications under Production Sharing Contract (PSC) regime for early monetisation of hydrocarbon discoveries
- Policy for extension of production sharing contracts
- Policy for early monetization of coal bed methane
- Setting up of national data repository
- Appraisal of un-appraised areas in sedimentary basins
- Re-assessment of hydrocarbon resources
- Policy framework to streamline the working of production sharing contracts in pre-NELP and NELP blocks
- Policy to incentivize enhanced recovery methods for oil and gas
- Policy framework for exploration and exploitation of unconventional hydrocarbons under existing production sharing contracts

29 Ministry of Coal, Extraction of Coal Bed Methane, 2023; Ministry of Petroleum and Natural Gas, Coal Bed Methane Policy, 2020

30 Ministry of Petroleum & Natural Gas, Cabinet approves revised domestic gas pricing guidelines, 2023

31 Ministry of Petroleum & Natural Gas, Demand and Supply of Natural Gas, 2019

32 Ministry of Petroleum & Natural Gas, PRANDHAN MANTRI UJJWALA YOJNA (PMUY), 2023

33 Ministry of Petroleum & Natural Gas, Import/Export of Oil, 2019

Way forward

This section highlights considerations for policymakers seeking to advance policy formulation and implementation in this sector. These are based on reports and materials prepared by key government bodies and expert institutions like the NITI Aayog and the Ministry of Petroleum and Natural Gas.

 Enhance exploration and diversify production	Intensify geological surveys and exploration - Evaluate and expand new areas for exploration (e.g., the opening up of the previously “no-go” Andaman and Nicobar Islands for exploration) Promote investments in new technologies and R&D - Invest in research in enhanced oil recovery techniques to extract hydrocarbons from existing and mature oilfields and enhancing their productive life	Further incentivize private and foreign investments in exploration and production, leveraging PPPs and financial incentives - In line with the Hydrocarbon Exploration and Licensing Policy, foster partnerships between public and private oil companies through joint exploration and production opportunities, risk-sharing PPP models (e.g., providing private firms incentives for investment in exploration blocks and infrastructure development) - Streamline regulatory approvals for exploration licenses and permits (for example by advancing the Open Acreage Licensing Policy), provide clear guidelines on taxation and royalty, and advance state-level single-window systems for approvals - Evaluate the use of tax incentives for exploration activities, profit-sharing mechanisms between public and private companies, and greater use of exploration subsidies for high-risk projects
 Boost efficiency and sustainability	Upgrade and modernize infrastructure to promote cleaner extraction technologies - Enhance the uptake and integration of renewable sources with oil operations, for example through the use of hybrid systems that deploy solar or wind power for pumping, refining, and transportation - Accelerate the use of digital technologies and AI for maintenance, modeling, and supply chain optimization efforts	Reduce carbon intensity in oil operations - Promote investments that enable greater biofuel blending, including through R&D in sustainable feedstocks for bioenergy production - Enhance demonstration projects in carbon capture, utilisation, and storage technologies (for example in enhanced oil recovery and sequestration) through financial incentives and viability-gap funding, market mechanisms like

	to enhance efficiency and cost-effectiveness	carbon pricing and markets, and public-private projects to demonstrate commercial scale
 Enable workforce transitions and integrated planning	Enable skills transitions in the oil and gas sector - Develop skills and livelihood transition assessments across upstream, downstream, and integrated businesses - Establish targeted reskilling (e.g., on integration of emerging technologies and digitalization of supply chains) and upskilling programs (e.g., for drilling engineers likely to see a decline in demand in skills) and ensure adequate diversity, equity, and inclusion of underrepresented segments (including in leadership positions)	Enable cross-sectoral and inter-ministerial collaboration and coordination - Define institutional structures (across states and between central and state ministries) to enable coordination for effective policymaking, including in matters such as the awarding exploration licenses, infrastructure development and consistency in licensing and safety norms, fuel pricing and subsidies, and the uptake of new green technologies

Recommended readings

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